Notes on Geometrization of real representations

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1 Analytic Rings

1.1 Light profinite sets

We started with the category of profinite sets, denote by Prof. The objects in Prof are thus functors $S : I \to$ **finSet** where I is cofiltered (NB: it is NOT a small category). There is a full-subcategory inside Prof, denoted by $\mathrm{Prof}_{\mathbb{N}}$, whose objects are functors $S : \mathbb{N} \to$ **finSet**, we call such functor a **light profinite set**.

Proposition 1.1. *The following categories (in the same color) are equivalent:*
(i) $\mathrm{Prof}_{\mathbb{N}} \subseteq \mathrm{Prof}$ *(ii) metrizable ones* $\subsetneq$ *totally disconnected compact Hausdorff spaces*
(iii) (countable Boolean algebras over $\mathbb{F}_2$*)*$^{\mathrm{op}} \subseteq$ *(Boolean algebras over* $\mathbb{F}_2$*)*$^{\mathrm{op}}$.

Proof. For $S \in \mathrm{Prof}$, we can write $S = \varinjlim_{i\in I} S_i$ and associate it with topological space $\varinjlim_{i\in I} S_i$ where $\varinjlim$ is taken in the category **Top**, this gives (i) $\Rightarrow$ (ii), denote also S the associated space. Then $C^0(S, \mathbb{F}_2)$ is a boolean algebra over $\mathbb{F}_2$, this gives (ii) $\Rightarrow$ (iii). For (iii) $\Rightarrow$ (ii), we take Spec A with Zariski topology for A Boolean algebra. For (ii) $\Rightarrow$ (i), we take the cofiltered system of finite quotient of S.
Restrict the above constructions to the corresponding subcategories gives the equivalences for $\mathrm{Prof}_{\mathbb{N}}$. ∎

Example 1.2. Here are some examples of light profinite sets:
(i) Any finite set $\in \mathrm{Prof}_{\mathbb{N}}$; (ii) $\mathbb{N} \cup \{\infty\} \in \mathrm{Prof}_{\mathbb{N}}$; (iii) The Cantor set $\{0,1\}^{\mathbb{N}} \in \mathrm{Prof}_{\mathbb{N}}$;
(iv) Any $S \in \mathrm{Prof}_{\mathbb{N}}$ admits a surjection $\{0,1\}^{\mathbb{N}} \twoheadrightarrow S$. ($\Rightarrow$ Any metrizable compact Hausdorff space X admits a surjection $\{0,1\}^{\mathbb{N}} \twoheadrightarrow X$.)

Proposition 1.3. *Any profinite set S is an injective object in* $\mathrm{Prof}_{\mathbb{N}}$.

Proof. Let $X \hookrightarrow Y$ be an injection with a map $X \to S$, we want to lift a map $Y \to S$. For the case when X, Y, S are all finite sets, it is OK. For arbitrary $X, Y \in \mathrm{Prof}_{\mathbb{N}}$ and finite S, we can consider the diagram:

$$\begin{tikzcd}
X \arrow[r] \arrow[d, hook] \arrow[rr, bend left] & S & X_i \arrow[l] \arrow[d] \\
Y \arrow[rr] & & Y_i \arrow[ul, dashed, "\exists"']
\end{tikzcd}$$

which reduce to the case of finite ones. For general case, we know there are maps $X \to S \to S_i$, where S_i finite, thus there is a map $f_i : Y \to S_i$. The question is reduced to whether f_{i+1} is a lift of f_i along $S_{i+1} \to S_i$. Now for $s \in S_i$ with a map $S_{i+1} \to \{s\}$, we can consider the base change $X_s \hookrightarrow Y_s$. We consider the following diagram

$$\begin{tikzcd}
X \arrow[r] & S_{i+1} \arrow[r] \arrow[dr] & S_i \\
X_s \arrow[u, hook] \arrow[r, hook] \arrow[rr, bend right] & Y_s \arrow[u, dashed, "\exists"] \arrow[r] & \{s\} \arrow[u, hook]
\end{tikzcd}$$

The map $Y_s \to S_{i+1}$ obtained from $X_s \to S_{i+1}$ is compatible with $S_{i+1} \to \{s\}$, i.e. all the closed loops in the diagram commute. ∎

1.2 Light condensed sets

The category $\mathrm{Prof}_{\mathbb{N}}$ of light profinite sets is endowed with Grothendieck topology generated by covers to be finite disjoint unions of jointly surjective maps, i.e. $\coprod_{i\in I} S_i \to S$, where I is a finite index set and $S = \cup_{i\in I} \mathrm{Im}(S_i)$.

Definition 1.4. Define the category of **light condensed sets** to be the category of sheaves $\mathrm{Prof}_{\mathbb{N}}^{\mathrm{op}} \to \mathbf{Sets}$, denoted by $\mathrm{CondSet}^{\mathrm{light}}$.

Let $F \in \mathrm{CondSet}^{\mathrm{light}}$, then we have $F(\emptyset) = \mathrm{pt}$; $F(S \sqcup T) = F(S) \times F(T)$ and finally for all $T \twoheadrightarrow S$, $F(S) = \mathrm{eq}(F(T) \rightrightarrows F(T \times_S T))$. There is a functor $\underline{\bullet} : \mathrm{Top} \to \mathrm{CondSet}^{\mathrm{light}} : X \to \underline{X}$, the later is defined by $\underline{X}(S) = C^0(S, X)$. In particular, when $S = \mathrm{pt}$, $\underline{X}(\mathrm{pt}) = X$ (as a set). If we take $S = \mathbb{N} \cup \{\infty\}$, then $\underline{X}(S)$ is the set of convergent sequences in X.
The functor $\underline{\bullet}$ has a left adjoint $\mathrm{CondSet}^{\mathrm{light}} \to \mathrm{Top} : X \mapsto X(*)_{\mathrm{top}}$. Here $X(*) = X(\mathrm{pt})$ is as above, serves as the underlying set of $X(*)_{\mathrm{top}}$, and the topology is given by the quotient topology from $\coprod_{S \in \mathrm{Prof}_{\mathbb{N}}} S \to X(*)$.

Proposition 1.5. *If X is a metrizable compact Hausdorff space, then the map $\underline{X}(*)_{\mathrm{top}} \to X$ is a homeomorphism.*

Proof. Assume $V \subseteq X$ and $\forall S \xrightarrow{p} X$, $p^{-1}(V)$ is closed, we want to show V is closed. In fact, for $x \in \overline{V}$ in the closure, $x = \varinjlim_n x_n, x_n \in V$ (sequentially closed). Then we get a map $p : \mathbb{N} \cup \{\infty\} \to X$ sending $n \mapsto x_n$ and $\infty \mapsto x$. We have $\mathbb{N} \subseteq p^{-1}(V)$, thus the later equals to $\mathbb{N} \cup \{\infty\}$, namely $x \in V$, which shows $V = \overline{V}$. ∎

- In $\mathrm{CondSet}^{\mathrm{light}}$, an object X is **quasi-compact (qc)** if there is a surjection $S \to X$ where $S \in \mathrm{Prof}_{\mathbb{N}}$ (Notice there is a Yoneda embedding: $\mathrm{Prof}_{\mathbb{N}} \to \mathrm{CondSet}^{\mathrm{light}}$).
- In $\mathrm{CondSet}^{\mathrm{light}}$, an object X is **quasi-separated (qs)** if $T \to X \leftarrow S$ where $T, S \in \mathrm{Prof}_{\mathbb{N}}$, then the fiber product $T \times_X S$ is quasi-compact.

Proposition 1.6. *(i) Assume $X \in \mathrm{CondSet}^{\mathrm{light}}$ is qcqs, then $X = \underline{X(*)_{\mathrm{top}}}$, where the corresponding $X(*)_{\mathrm{top}}$ is a metrizable compact Hausdorff space. (Hence $\mathrm{Prof}_{\mathbb{N}}$ are qcqs objects in $\mathrm{CondSet}^{\mathrm{light}}$)*
(ii) If X is quasi-separated, then $X = \varinjlim_i X_i$ is a (filtered) colimit of qcqs ones.

Proof. Since X is qc, let $\mathrm{Prof}_{\mathbb{N}} \ni S \twoheadrightarrow X$ be the surjection from light profinite set. Since X is qs, $S \times_X S$ is qc. Thus there exists $\mathrm{Prof}_{\mathbb{N}} \ni T \twoheadrightarrow S \times_X S$. We also have $X \to \underline{X(*)_{\mathrm{top}}}$ by adjunction of functors. We get

$$\begin{tikzcd} T \arrow[r, shift left] \arrow[r, shift right] \arrow[d, equal] & S \arrow[r, two heads] \arrow[d, equal] & X \arrow[d] \\ T \arrow[r, shift left] \arrow[r, shift right] & S \arrow[r, two heads] & \underline{X(*)_{\mathrm{top}}} \end{tikzcd}$$

The image $\mathrm{Im}(T \subseteq S \times S)$ is a compact equivalence relation in $\mathrm{Prof}_{\mathbb{N}}$ so that X equals to $S \times S$ quotient by this relation and that $X(*)_{\mathrm{top}}$ is metrizable compact Hausdorff space.
For (ii) we can consider $\coprod_{f \in X(S)} S \xrightarrow{f} X$. In fact, one can take $S = \{0,1\}^{\mathbb{N}}$ to be the Cantor set. The image T of f is qcqs as $S \times_T S = S \times_X S$ is light profinite. ∎

Remark 1.7. As is mentioned in [SC, Episode 3/24], it is important to let the test category $\mathrm{Prof}_{\mathbb{N}}$ be finitary and contains the Cantor set. If one only allows $\mathbb{N} \cup \{\infty\}$, then the quasi-compact objects will all be countable.

1.3 Light condensed abelian groups

Definition 1.8. We can define the category of **Light condensed abelian groups** to be abelian sheaves in $\mathrm{CondSet}^{\mathrm{light}}$, denoted by $\mathrm{CondAb}^{\mathrm{light}}$.

The natural forgetful functor $\mathrm{CondAb}^{\mathrm{light}} \to \mathrm{CondSet}^{\mathrm{light}}$ has a left adjoint functor:

$$\mathrm{CondSet}^{\mathrm{light}} \to \mathrm{CondAb}^{\mathrm{light}}, X \mapsto \mathbb{Z}[X]$$

where $\mathbb{Z}[X]$ is the sheafification of the functor $S \mapsto \mathbb{Z}[X(S)]$. One can also say the tensor product in $\mathrm{CondAb}^{\mathrm{light}}$. Likewise, $X \otimes_{\mathbb{Z}} Y$ is the sheafification of $X(S) \otimes_{\mathbb{Z}} Y(S)$.

Remark 1.9. Notice that for different topological structures on one abelian group, the corresponding light condensed abelian groups could be different. For example, both $\underline{\mathbb{R}}$ and $\underline{\mathbb{R}_{\mathrm{disc}}}$ are light condensed abelian groups. $\underline{\mathbb{R}}/\underline{\mathbb{R}_{\mathrm{disc}}}(S) = C^0(S,\mathbb{R})/C^{\mathrm{loc.cons}}(S,\mathbb{R})$ could be large when one take S to be the Cantor set.

Proposition 1.10. *The category* $\mathrm{CondAb}^{\mathrm{light}}$ *is a Grothendieck abelian category, namely abelian category with all colimits, filtered colimits being exact and with generator.*

Proof. The generator is $\mathbb{Z}\left[\{0,1\}^{\mathbb{N}}\right]$. The other properties are not difficult to see except that we need countable products to be exact. This comes from the property that sequential limits of surjective maps are surjective. In fact, given $f_n : M_n \twoheadrightarrow N_n$ surjective, we want $\prod_n f_n : \prod_n M_n \to \prod_n N_n$ to be surjective. This map can be written as $\varprojlim_m \left(\prod_{n\leqslant m} M_n \times \prod_{n>m} M_n \to \prod_n N_n\right)$. Now for $M = \varprojlim_n M_n \to M_0$ in $\mathrm{CondAb}^{\mathrm{light}}$ being surjective, consider $S_0 \to M_0$ where $S \in \mathrm{Prof}_{\mathbb{N}}$ (generators of the site), we need lift $S_0 \to M$. But it is enough to find $S \twoheadrightarrow S_0$ surjection with a map $S \to M$. We start with $S_0 \to M_0$, then there is a surjection $S_1 \twoheadrightarrow S_0$ that lift to $S_1 \to M_1$. Take the limit we get $S := \varprojlim_n S_n \to S_0$. And we conclude with the following easy-verify fact that sequential limit of surjective maps in $\mathrm{Prof}_{\mathbb{N}}$ is again surjective. ■

Remark 1.11. The point that countable limit of covers is still a cover forces totally disconnected spaces to be the building blocks in the site. For example, consider the limit of covers of the interval gives Cantor set.

We have internal extension in $\mathrm{CondAb}^{\mathrm{light}}$. More concretely it is given by $\underline{\mathrm{Ext}}^i(X,Y)(S) = \mathrm{Ext}^i(X \otimes \mathbb{Z}[S], Y)$.

Proposition 1.12. *In* $\mathrm{CondAb}^{\mathrm{light}}$, $\mathbb{Z}[\mathbb{N}\cup\{\infty\}]$ *is internally projective, i.e.* $\underline{\mathrm{Ext}}^i\left(\mathbb{Z}[\mathbb{N}\cup\{\infty\}], \bullet\right) = 0, i > 0$.

Proof. Let $P := \mathbb{Z}[\mathbb{N}\cup\{\infty\}]/\mathbb{Z}[\infty]$, then P classifies the null sequences. Want to show that P is internally projective. It is enough to show for any $S \in \mathrm{Prof}_{\mathbb{N}}$ and $p : M \twoheadrightarrow N$ surjection in $\mathrm{CondAb}^{\mathrm{light}}$ with a map $g : \mathbb{Z}[S] \otimes P \to N$, there exists a lift $\mathbb{Z}[S] \otimes P \to M$. Now by construction, g is a map $S \times (\mathbb{N}\cup\{\infty\})$ that sends $S \times \{\infty\}$ to 0. There exists $g' : S' \in \mathrm{Prof}_{\mathbb{N}} \twoheadrightarrow S \times (\mathbb{N}\cup\{\infty\})$ that lifts g (essentially the definition of surjectivity in $\mathrm{CondAb}^{\mathrm{light}}$). There exists $S'' \subseteq S'$ closed so that $S'' \times_{\mathbb{N}\cup\{\infty\}} \mathbb{N} \cong S \times \mathbb{N}$. Let $S'_\infty := S' \times_{S\times(\mathbb{N}\cup\{\infty\})} S \times \{\infty\}$ with injection $i : S'_\infty \hookrightarrow S'$. There exists a retraction $r : S' \to S'_\infty$ due to the following Lemma:

Lemma 1.13. *Every object in* $\mathrm{Prof}_{\mathbb{N}}$ *is injective in* $\mathrm{Prof}_{\mathbb{N}}$.

Now consider $g'' := g' - g' \circ (r \circ i) : \mathbb{Z}[S'] \to M$ that factors through $\mathbb{Z}[S']/\mathbb{Z}[S'_\infty] \cong \mathbb{Z}[S] \otimes P$, we get the desired lift. ■

Remark 1.14. Note that if we take $S = *$, then we can see internally projective implies projective. Thus P and $\mathbb{Z}[\mathbb{N} \cup \{\infty\}]$ are projective objects in $\mathrm{CondAb}^{\mathrm{light}}$. However, $\mathbb{N} \cup \{\infty\}$ projective in $\mathrm{Prof}_{\mathbb{N}}$ or $\mathrm{CondSet}^{\mathrm{light}}$. For example if we consider the cover $M = (2\mathbb{N} \cup \{\infty\}) \cup ((2\mathbb{N} + 1) \cup \{\infty\}) \twoheadrightarrow N = \mathbb{N} \cup \{\infty\}$, there is no lift of id_N to M. Also, if we remove the 'light' condition, then $\mathbb{Z}[\mathbb{N} \cup \{\infty\}]$ is not projective in CondAb.

1.4 Analytic rings

We start with $R^{\triangleright}$ a light condensed commutative ring, i.e. a sheaf $\mathrm{Prof}_{\mathbb{N}}^{\mathrm{op}} \to \mathrm{Ring}$ and we write $\mathrm{Mod}_{R^{\triangleright}}$ modules over $R^{\triangleright}$ in $\mathrm{CondAb}^{\mathrm{light}}$.

Definition 1.15. An **analytic ring** is a pair $R := \left(R^{\triangleright}, \mathrm{Mod}_R \subseteq \mathrm{Mod}_{R^{\triangleright}}\right)$ such that Mod_R is a full subcategory that stable under limit, colimit and extension. Moreover, the internal extension $\underline{\mathrm{Ext}}^i_{\mathrm{Mod}_{R^{\triangleright}}}(M, N) \in \mathrm{Mod}_R$ whenever $M \in \mathrm{Mod}_{R^{\triangleright}}, N \in \mathrm{Mod}_R$. And finally we require $R^{\triangleright} \in \mathrm{Mod}_R$.

There is a left adjoint of the inclusion functor:

$$(\bullet)^{\square} : \mathrm{Mod}_{R^{\triangleright}} \to \mathrm{Mod}_R : M \mapsto M^{\square} = M \otimes_{R^{\triangleright}} R \quad \text{(both are just notations!)}$$

In fact for any $M \in \mathrm{Mod}_{R^{\triangleright}}$, we can consider the product $M \xrightarrow{p} \prod_{M \twoheadrightarrow M' \in \mathrm{Mod}_R} M'$ and manually set

$$M^{\square} := \left\{ p(m) \in \prod M' \mid \forall f \in \mathrm{End}(\prod M'), f(p(m)) = p(m) \right\}$$

The kernel of $(\bullet)^{\square}$ is a symmetric monoidal ideal, namely if $M \in \mathrm{Mod}_{R^{\triangleright}}$ with $M^{\square} = 0$, then $(N \otimes M)^{\square} = 0$ for all $N \in \mathrm{Mod}_{R^{\triangleright}}$. In fact, let $L \in \mathrm{Mod}_R$ be arbitrary test object, we have:

$$\begin{aligned} \mathrm{Hom}_{\mathrm{Mod}_{R^{\triangleright}}}(N \otimes_{R^{\triangleright}} M, L) &= \mathrm{Hom}_{\mathrm{Mod}_{R^{\triangleright}}}(M, \underline{\mathrm{Hom}}_{\mathrm{Mod}_{R^{\triangleright}}}(N, L)) \quad \underline{\mathrm{Hom}}_{\mathrm{Mod}_{R^{\triangleright}}}(N, L) \in \mathrm{Mod}_R \text{ by definition.} \\ &= \mathrm{Hom}_{\mathrm{Mod}_R}(M^{\square}, \underline{\mathrm{Hom}}_{\mathrm{Mod}_{R^{\triangleright}}}(N, L)) = 0 \end{aligned}$$

We also have a unique symmetric monoidal structure in Mod_R so that $(\bullet)^{\square}$ becomes symmetric monoidal functor. For $M, N \in \mathrm{Mod}_R$, define $M \otimes_R N := (M \otimes_{R^{\triangleright}} N)^{\square}$. To check if symmetric monoidal, for $M, N \in \mathrm{Mod}_{R^{\triangleright}}$, we want to see $= (M^{\square} \otimes_{R^{\triangleright}} N^{\square})^{\square} = \left(M^{\square} \otimes_{R^{\triangleright}} N^{\square}\right)^{\square} =: M^{\square} \otimes_R N^{\square}$. The proof goes in the similar way as solidification functor case: it is enough to show that $(M \otimes_{R^{\triangleright}} N)^{\square} \cong (M^{\square} \otimes_{R^{\triangleright}} N)^{\square}$. For arbitrary $S \in \mathrm{Mod}_R$

$$\begin{aligned} \mathrm{Hom}_{\mathrm{Mod}_{R^{\triangleright}}}\left(M^{\square} \otimes_{R^{\triangleright}} N, S\right) &= \mathrm{Hom}_{\mathrm{Mod}_{R^{\triangleright}}}\left(N, \underline{\mathrm{Hom}}_{\mathrm{Mod}_{R^{\triangleright}}}(M^{\square}, S)\right) = \mathrm{Hom}_{\mathrm{Mod}_R}(N^{\square}, \underline{\mathrm{Hom}}_{\mathrm{Mod}_R}(M^{\square}, S)) \\ &= \mathrm{Hom}_{\mathrm{Mod}_R}(M^{\square} \otimes_R N^{\square}, S) = \mathrm{Hom}_{\mathrm{Mod}_{R^{\triangleright}}}\left(M^{\square} \otimes_{R^{\triangleright}} N^{\square}, S\right) \end{aligned}$$

Remark 1.16. Without the last condition that $R^{\triangleright} \in \mathrm{Mod}_R$, then R is called a **pre-analytic ring**.

Example 1.17 (**Solid abelian groups**)**.** Now we consider the analytic ring Solid with the corresponding $R^{\triangleright} = \mathbb{Z}$ and the subcategory Solid $\subseteq \mathrm{CondAb}^{\mathrm{light}} = \mathrm{Mod}_{\mathbb{Z}}$ that makes $(\mathbb{Z}, \mathrm{Solid}(\mathbb{Z}))$ an analytic ring.
Let shift $: \mathbb{N} \cup \{\infty\} \to \mathbb{N} \cup \{\infty\}$ be the map $n \mapsto n+1$ and $\infty \mapsto \infty$, it is also an endomorphism over P. Define:

Definition 1.18. For $M \in \mathrm{CondAb}^{\mathrm{light}}$, call M **solid** if $\underline{\mathrm{Hom}}(P, M) \xrightarrow{f=\mathrm{Id}-\mathrm{Shift}} \underline{\mathrm{Hom}}(P, M)$ is an isomorphism.

An intuition for the above condition is that we can regard $\mathrm{Hom}(P, M)$ as the collection of null sequences in M. Let $(m_n)_{n\in\mathbb{N}}$ be a null sequence, then $f = \mathrm{Id} - \mathrm{Shift}$ maps it to $(f(m)_n) := (m_n - m_{n+1})_{n\in\mathbb{N}}$. Then f has a formal inverse "f^{-1}" that formally in the form $f^{-1}(f_n)\varepsilon = \varepsilon(\sum_{n=0}^{\infty} f_n, \sum_{n=1}^{\infty} f_n, \ldots)$. For requiring f to be a bijection then amounts to say the inverse do exist, namely null sequences need to be summable.

Theorem 1.19. *Let* Solid *stands for solid light abelian groups, then* $(\mathbb{Z}, \mathrm{Solid})$ *is an analytic ring.*

Proof. The internal Hom preserves limits, also P is a compact object thus preserves colimits. Also since P is internally projective so we do not need pass to derived sense. For $M \in \mathrm{CondAb}^{\mathrm{light}}$ and $N \in \mathrm{Solid}$, we want to see if $\underline{\mathrm{Hom}}(M, N) \in \mathrm{Solid}$, we have $\underline{\mathrm{Hom}}(P, \underline{\mathrm{Hom}}(M, N)) \cong \underline{\mathrm{Hom}}(P \otimes M, N) \cong \underline{\mathrm{Hom}}(M, \underline{\mathrm{Hom}}(P, N))$. Thus we can see the induced endomorphism by $f = \mathrm{Id} - \mathrm{shift}$ is an isomorphism, due to the fact that $N \in \mathrm{Solid}$. This also applies to higher internal extensions i.e. $\underline{\mathrm{Hom}}(P, \underline{\mathrm{Ext}}^i(M, N)) = \underline{\mathrm{Hom}}(M, \underline{\mathrm{Ext}}^i(P, N))$.
Finally, we find that $\underline{\mathrm{Hom}}(P, \mathbb{Z}) = \bigoplus_{\mathbb{N}} \mathbb{Z}$, on which f acts as an isomorphism, thus $\mathbb{Z} \in \mathrm{Solid}$. ∎

As an conclusion, we get the left adjoint functor $(\bullet)^{\square} : \mathrm{CondAb}^{\mathrm{light}} \to \mathrm{Solid}$, this suggests the name **solidification functor**.

Theorem 1.20. *For* $S \in \mathrm{Prof}_{\mathbb{N}}$ *an infinite light profinite set, there exists a morphism* $P \to S$ *so that the induced solidification map* $P^{\square} \to \mathbb{Z}[S]^{\square}$ *is an isomorphism. Moreover,* $P^{\square}$ *is a compact projective generator in* Solid.

Proof. We first construct Denote $\pi_n : S \to S_n$ the projection map. For any sequence of compatible lifts $S_0 \xrightarrow{J_0} S_1 \xrightarrow{J_1} S_2 \to \ldots \to S$, write $\iota_n : S_n \to S$. The collection $\cup_n \iota_n(S_n) \cong \mathbb{N}$, thus we get a map $\mathbb{N} \to S$. For any $a \in \cup_n \iota_n(S_n)$, the sequence $(\iota_n(a) - \iota_{n-1}(a))_{n\in\mathbb{N}}$ is a null sequence in $\mathbb{Z}[S]$ and gives an injective map $g : P \to \mathbb{Z}[S]$. We consider the diagram in the following left:

$$\begin{tikzcd} P \otimes \mathbb{Z}[S] \arrow[r, "G"] \arrow[d, "f\otimes\mathrm{id}"'] & P \arrow[d, "g"] \\ P \otimes \mathbb{Z}[S] \arrow[r, "F"] & \mathbb{Z}[S] \arrow[l, bend left, dashed] \end{tikzcd} \qquad \begin{tikzcd} (P \otimes \mathbb{Z}[S])^{\square} \arrow[r, "G^{\square}"] \arrow[d, "\cong"'] & P^{\square} \arrow[d] \\ (P \otimes \mathbb{Z}[S])^{\square} \arrow[r, "F^{\square}"] & \mathbb{Z}[S]^{\square} \arrow[l, bend left, dashed] \end{tikzcd}$$

We let $F = (\sigma_n)_{n=0}^{\infty}$ so that σ_n is the restriction $F : P \to \mathrm{End}(\mathbb{Z}[S])$ to $\{n\}$. Concretely $\{n\} \times S \to \mathbb{Z}[S]$ is given by $\mathrm{id}_{\mathbb{Z}[S]} - \iota_{n-1} \circ \pi_{n-1}$ (we take $\iota_{-1} \circ \pi_{-1}$ to be zero for convenience). The composition $F \circ (f \otimes \mathrm{id})$ is thus given by $(\sigma_n - \sigma_{n+1})_{n=0}^{\infty}$. Note that $\sigma_n - \sigma_{n+1} = \iota_n \circ \pi_n - \iota_{n-1} \circ \pi_{n-1}$ and thus factors P via $P \xrightarrow{g} \mathbb{Z}[S]$. Pass to the solidification, we have the right above diagram. The dotted map is given by the inclusion $\mathbb{Z}\{0\} \otimes \mathbb{Z}[S] \hookrightarrow P \otimes \mathbb{Z}[S]$ then $\mathbb{Z}[S]^{\square}$ is a retract of $P^{\square}$. The composition $r : P^{\square} \to \mathbb{Z}[S]^{\square} \to P^{\square}$ is thus idempotent. We want r to be identity, which thus shows $P^{\square} \cong \mathbb{Z}[S]^{\square}$. In fact, we can replace in the above two diagrams $\mathbb{Z}[S]$ to P, i.e. we restrict the maps F and G from $\mathbb{Z}[S]$ to P then the idempotent $r : P^{\square} \to \mathbb{Z}[S]^{\square} \to P^{\square}$ is the same as $P^{\square} \to P^{\square} \to P^{\square}$. The map

corresponding g is nothing but id_P (in that case $\iota_n = \mathrm{id}_{\mathbb{N}}$). We get $r = \mathrm{id}_{P^\square}$ as desired.
We leave the calculation of $P^\square$ to the sequel. But we can immediately notice that as a corollary of the above argument, $P^\square \cong \mathbb{Z}\left[\mathbb{N} \cup \{\infty\}\right]^\square \cong \mathbb{Z}\left[\{0,1\}^{\mathbb{N}}\right]^\square$ is a projective generator à la fois. ∎

Remark 1.21. There is an isomorphism between light condensed abelian groups P and $\mathbb{Z}[\hat{T}]$, where $\mathbb{Z}[\hat{T}]$ is defined to be $\cup_{n\geqslant 0}\left(\varprojlim_i (\mathbb{Z}[T]/T^i)_{\ell^1\leqslant n}\right)$ and the isomorphism is given by $n \mapsto T^n$. Indeed, in the diagram of aobve proof, if we take $\mathbb{Z}[S] = \mathbb{Z}[\hat{T}]$, then σ_n is in fact the endomorphism $a = \sum_{k=0}^{\infty} a_k T^k \mapsto \sum_{k=n}^{\infty} a_k T^k$. And the map $G : P \otimes \mathbb{Z}[\hat{T}] \to P$ sends $\{n\} \otimes a \mapsto a_n$.

Lemma 1.22. *The solidification $\mathbb{R}^\square$ of $\mathbb{R}$ is zero.*

Proof. Let $g : P \to \mathbb{R}$ be any null sequence in $\mathbb{R}$. There exist a map $h_g : P \to \mathbb{R}^\square$ such that the diagram

$$\begin{tikzcd} P \arrow[r, "g"] \arrow[d, "f=\mathrm{Id}-\mathrm{shift}"'] & \mathbb{R} \arrow[d] \\ P \arrow[r, "h_g"] & \mathbb{R}^\square \end{tikzcd}$$

commutes, where $\mathbb{R} \to \mathbb{R}^\square$ is the right adjoint of $\mathrm{Id}_{R^\square}$. Let $\rho : \mathbb{Z} \to P : 1 \mapsto T^0$, we denote by $c_g := h_g \circ \rho(1)$. Note that $h_g(\rho(1)) = h_g(T^0) = h_g(f(\sum_{i=0}^{\infty} T^i))$. we denote $g^\square := h_g \circ f$. Let also $\alpha : P \to P$ given by $[n] \mapsto [2n] + [2n+1], \infty \mapsto [0]$. Start with g_0 a null sequence given by

$$g_0 = \left(\frac{1}{2}, \frac{1}{2}, \ldots, \underbrace{\frac{1}{2^n}, \ldots, \frac{1}{2^n}}_{2^n\text{-terms}}, \ldots \right)$$

Then we have $g_0 \circ \alpha = (1, \frac{1}{2}, \frac{1}{2}, \ldots) = g_0 \circ \mathrm{shift} + (1, 0, 0, \ldots)$. The corresponding $c_{g_0\circ\alpha} = c_{g_0\circ\mathrm{shift}} + 1$, since $\mathrm{shift}(\sum_{n=0}^{\infty} T^n) = \sum_{n=0}^{\infty} T^n$, thus $c_{g_0\circ\alpha} = c_{g_0} + 1$. On the other hand, we can calculate directly

$$c_{g_0\circ\alpha} = g_0^\square\left(\sum_{n=0}^{\infty}(T^{2n} + T^{2n+1})\right) = g_0^\square\left(\sum_{n=0}^{\infty} T^n\right) = c_{g_0}$$

Thus we get $c_{g_0} + 1 = c_{g_0}$, namely $0 = 1$ in $\mathbb{R}^\square$, which forces $\mathbb{R}^\square = 0$. ∎

Theorem 1.23. *We have $P^\square \cong \prod_{\mathbb{N}} \mathbb{Z} \cong \mathbb{Z}[\![T]\!]$. Moreover, for any infinite light profinite group S, we have $\mathbb{Z}[S]^\square \cong \varprojlim_i \mathbb{Z}[S_i] \cong \prod_{\mathbb{N}} \mathbb{Z}$.*

Proof. We consider $\prod_{\mathbb{N}}^{\mathrm{bnd}} \mathbb{Z} := \cup_{n\in\mathbb{N}} \left(\prod_{\mathbb{N}}(\mathbb{Z} \cap [-n,n])\right)$ and $\prod_{\mathbb{N}}^{\mathrm{bnd}} \mathbb{R} := \cup_{n\in\mathbb{N}} \left(\prod_{\mathbb{N}}(\mathbb{R} \cap [-n,n])\right)$ the bounded product of $\mathbb{Z}$ (resp. $\mathbb{R}$)'s. We have exact sequences:

$$0 \longrightarrow \prod_{\mathbb{N}} \mathbb{Z} \longrightarrow \prod_{\mathbb{N}} \mathbb{R} \longrightarrow \prod_{\mathbb{N}} \mathbb{R}/\mathbb{Z} \longrightarrow 0; \quad 0 \longrightarrow \prod_{\mathbb{N}}^{\mathrm{bnd}} \mathbb{Z} \longrightarrow \prod_{\mathbb{N}}^{\mathrm{bnd}} \mathbb{R} \longrightarrow \prod_{\mathbb{N}} \mathbb{R}/\mathbb{Z} \longrightarrow 0$$

which shows $\prod_{\mathbb{N}} \mathbb{Z}/\prod_{\mathbb{N}}^{\mathrm{bnd}} \mathbb{Z} \cong \prod_{\mathbb{N}} \mathbb{R}/\prod_{\mathbb{N}}^{\mathrm{bnd}} \mathbb{R}$. Hence from Lemma 1.22, we get $(\prod_{\mathbb{N}}^{\mathrm{bnd}} \mathbb{Z})^{\square} \cong (\prod_{\mathbb{N}} \mathbb{Z})^{\square}$. The solidification of the later one is itself as one can verify that $\prod_{\mathbb{N}} \mathbb{Z} \in \mathrm{Solid}$ by definition. Finally, one can show that there exists a natural isomorphism between $P^{\square}$ and $(\prod_{\mathbb{N}}^{\mathrm{bnd}} \mathbb{Z})^{\square}$ with similar way as in Theorem 1.20. ■

Example 1.24 (Some examples of solidification)**.** (i) We write $\otimes_{\mathbb{Z}_{\square}}$ the tensor product in Solid, also $\overset{\mathbb{L}}{\otimes}_{\mathbb{Z}_{\square}}$ for the derived version in D(Solid). Then $\mathbb{Z}[\![T]\!] \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}_{\square}} \mathbb{Z}[\![U]\!] = \mathbb{Z}[\![T, U]\!]$. The later one is isomorphic to $\mathbb{Z}[\![T']\!]$ via re-arranging of $\mathbb{N}$.
(ii) The p-adic integers $\mathbb{Z}_p$ is itself a solid abelian group. We can write $\mathbb{Z}_p = \mathbb{Z}[\![T]\!]/(T-p)$, then

$$\mathbb{Z}_p \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}_{\square}} \mathbb{Z}_l = \begin{cases} 0 & l \neq p \\ \mathbb{Z}_p & l = p \end{cases}$$

In fact we have $\mathbb{Z}_p \otimes_{\mathbb{Z}_{\square}} \mathbb{Z}_l = \mathbb{Z}_p \otimes_{\mathbb{Z}_{\square}} (\mathbb{Z}[\![U]\!]/(U-l)) \cong \mathbb{Z}_p[\![U]\!]/(U-l)$. One can see if $l \neq p$, $U - l$ is an invertible power series.
(iii) Let $\mathbb{Z}_{p,\square}$ be the analytic ring $(\mathbb{Z}_p, \mathrm{Mod}_{\mathbb{Z}_p}(\mathrm{Solid}))$. Consider $\hat{\bigoplus}_{n\in\mathbb{N}} \mathbb{Z}_p$ the p-completion of $\bigoplus_{n\in\mathbb{N}} \mathbb{Z}_p$. $\hat{\bigoplus}_{n\in\mathbb{N}} \mathbb{Z}_p \cong \varprojlim_{i\in\mathbb{N}} \left(\bigoplus_{n\in\mathbb{N}} \mathbb{Z}_{p^i}\right)$, then we have

$$\hat{\bigoplus}_{n\in\mathbb{N}} \mathbb{Z}_p \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}_{p,\square}} \hat{\bigoplus}_{m\in\mathbb{N}} \mathbb{Z}_p \cong \hat{\bigoplus}_{n,m\in\mathbb{N}} \mathbb{Z}_p$$

Proof. We can write $\hat{\bigoplus}_{n\in\mathbb{N}} \mathbb{Z}_p \cong \varinjlim_{f:\mathbb{N}\to\mathbb{N}} \prod_{n\in\mathbb{N}} p^{f(n)}\mathbb{Z}_p$, where in the colimit f are taken so that $f(n)$ goes to ∞ when $n \to \infty$. Now $\prod_{n\in\mathbb{N}} p^{f(n)}\mathbb{Z}_p \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}_{p,\square}} \prod_{n\in\mathbb{N}} p^{g(m)}\mathbb{Z}_p = \prod_{n,m\in\mathbb{N}} p^{f(n)+g(m)}\mathbb{Z}_p$. ■

We have seen that the solidification of $\mathbb{R}$ is less interesting. To work with real analytic stuffs, one need pass to another 'phase' gas. Intuitively, let q be a formal variable, for $M \in \mathrm{Mod}_{\mathbb{Z}[q]}$, consider the induced endomorphism of $\mathrm{Id} - q \cdot \mathrm{shift}$ on $\underline{\mathrm{Hom}}(P \otimes_{\mathbb{Z}} \mathbb{Z}[q], M)$ that sends $(m_0, m_1, \ldots) \mapsto (m_0 - qm_1, m_1 - qm_2, \ldots)$. The formal inverse is given by $(n_0, n_1, \ldots) \mapsto (\sum_{i=0}^{\infty} q^i n_i, \sum_{i=1}^{\infty} q^{i-1} n_i, \ldots)$. Similar to the requirement for Solid, we want this inverse to exist. In particular, we define the gaseous analytic ring to be

Definition 1.25. Define $\mathbb{R}^{\mathrm{gas}} = (\mathbb{R}, \mathrm{Mod}_{\mathbb{R}^{\mathrm{gas}}})$ the **gaseous real analytic ring** with

$$\mathrm{Mod}_{\mathbb{R}^{\mathrm{gas}}} = \left\{ M \in \mathrm{Mod}_{\mathbb{R}} \mid \underline{\mathrm{Hom}}(P, M) \xrightarrow{\mathrm{Id} - \frac{1}{2}\cdot \mathrm{shift}} \underline{\mathrm{Hom}}(P, M) \text{ isomorphism} \right\}$$

More generally, $M \in \mathrm{Mod}_{\mathbb{Z}[q]^{\mathrm{gas}}}$ if $\mathrm{Id} - q \cdot \mathrm{shift}$ acts on $\underline{\mathrm{Hom}}(P, M)$ as an isomorphism.

Remark 1.26. The condition $M \in \mathrm{Mod}_{\mathbb{R}^{\mathrm{gas}}}$ is equivalent to the condition that $\mathrm{Id} - a \cdot \mathrm{shift}$ acts on $\underline{\mathrm{Hom}}(P, M)$ as an isomorphism, for any $a \in (0, 1)$. We call a module satisfies this condition to be a-gaseous.

Lemma 1.27. *Let V be the topological inverse limit of Banach space over $\mathbb{R}$ along continuous $\mathbb{R}$-linear maps, then $\underline{V}$ is a-gaseous for any $a \in [0, 1)$.*

Proof. Recall $\underline{\bullet}$ is the functor Top $\to$ CondSet$^{\text{light}}$ that right adjoint to the functor CondSet$^{\text{light}}$ that commutes with inverse limit, so we only need to consider for one Banach space V. For any $S \in \text{Prof}_{\mathbb{N}}$, we have:

$$\begin{aligned}&\underline{\text{Hom}}_{\text{CondAb}^{\text{light}}}(\mathbb{Z}[\mathbb{N} \cup \{\infty\}], \underline{V})(S) = \text{Hom}_{\text{CondAb}^{\text{light}}}(\mathbb{Z}[S \times (\mathbb{N} \times \{\infty\}), \underline{V}]) \\ =&\mathcal{C}_0(S \times (\mathbb{N} \times \{\infty\}), V) = \mathcal{C}_0(S, \mathcal{C}_0(\mathbb{N} \cup \{\infty\}), V) = \underline{\mathcal{C}_0(\mathbb{N} \cup \{\infty\}, V)}(S)\end{aligned}$$

Thus $\underline{\text{Hom}}_{\text{CondAb}^{\text{light}}}(\mathbb{Z}[\mathbb{N} \cup \{\infty\}], \underline{V}) \cong \underline{\mathcal{C}_0(\mathbb{N} \cup \{\infty\}, V)}$. For a null sequence $(x_0, x_1, ...)$ in Banach space V, we want the sequence $(y_0, y_1, ...) := \left(\sum_{i=0}^{\infty} a^i x_i, \sum_{i=1}^{\infty} a^{i-1} x_i, ...\right)$ to be a null sequence for any $a \in [0, 1)$. For any $\delta > 0$, there exists $N \in \mathbb{N}$ so that $\forall n \geqslant N$, $|x_n| < \frac{\delta}{2}$, thus $y_n = \sum_{k=n}^{\infty} a^{k-n} x_k < \delta$, as we want. ■

Lemma 1.28. *Let $M \in \text{Mod}_{\mathbb{Z}[q]}$ is q-gaseous if and only if M is q^n-gaseous.*

Proof. For $(a_0, a_1, ...)$ a null sequence, the formal sum $\sum_{i=0}^{\infty} a_i (q^n)^i = a_0 + 0 \cdot q + ... + 0 \cdot q^{n-1} + a_1 q^n + 0 \cdot q^{n+1} + ...$, thus we can construct another null sequence $(a_0, 0, ..., 0, a_1, ...)$ in M by inserting zero's. Assume M is q-gaseous, then its q-associated formal sum exists, which equals to the q^n-associated formal sum of the original null sequence $(a_0, a_1, ...)$, which shows M is q^n-gaseous.
Conversely, $\sum_{i=0}^{\infty} a_i q^i = \sum_{j=0}^{n-1} q^j \cdot (\sum_{i=0}^{\infty} a_{j+ni} q^{ni})$, which shows if M is q^n-gaseous, then it is also q-gaseous. ■

Remark 1.29. One can similarly define $\mathbb{Q}_{p,\text{gas}} = (\mathbb{Q}_p, \text{Mod}_{\mathbb{Q}_p,\text{gas}})$. However for V, W Banach spaces over $\mathbb{Q}_p$, $\underline{V} \otimes_{\mathbb{Q}_{p,\text{gas}}} \underline{W} \neq \underline{V \hat{\otimes} W}$.

Back to cases over $\mathbb{R}$, in which we work with the overconvergent algebras

$$A(T_1, ..., T_d)^t := \left\{ \sum_{n_1,...,n_d \geqslant 0} a_{n_1,...,n_d} T^{n_1} ... T^{n_d} \;\middle|\; \begin{array}{c} a_{n_1,...,n_d} \in \mathbb{R}, \exists r > 1 \text{ such that} \\ r^{\sum_{i=1}^d n_i} \left|a_{n_1,...,n_d}\right| \to 0 \end{array} \right\}$$

When there is one variable, we see that $A(T)^t = \varinjlim_{r>1} (A_r(T) = \underline{\mathcal{C}_0(\mathbb{N}, \mathbb{R})})$, where $\underline{\mathcal{C}_0(\mathbb{N}, \mathbb{R})} = \underline{\text{Hom}}(P, \underline{\mathbb{R}})$. In particular, $A(T)^t \in \text{Mod}_{\mathbb{R}^{\text{gas}}}$. For $1 < r' < r$, the injective transition map in the above colimit is given by $A_r(T) \xrightarrow{(\cdot (\frac{r'}{r})^n)_{n \in \mathbb{N}}} A_{r'}(T)$. Thus, $A_r(T) \to A(T)^t$ is given by $(b_n)_{n \in \mathbb{N}} \mapsto \sum_{n=0}^{\infty} \frac{b_n}{r^n} T^n$. We want to see if

$$A(T_1)^t \overset{\mathbb{L}}{\otimes}_{\mathbb{R}^{\text{gas}}} A(T_2)^t \cong A(T_1, T_2)^t = \varinjlim \mathcal{C}_0(\mathbb{N} \times \mathbb{N}, \mathbb{R})$$

We set $P^{\text{gas}} := P \otimes_{\mathbb{Z}} \mathbb{R}^{\text{gas}}$, namely P^{gas} is the analytification of $P \otimes_{\mathbb{Z}} \mathbb{R}$. We have $\underline{\text{Hom}}_{\mathbb{R}}(P^{\text{gas}}, \underline{\mathbb{R}}) \cong \underline{\text{Hom}}_{\mathbb{Z}}(P, \underline{\mathbb{R}}) = \underline{\mathcal{C}_0(\mathbb{N}, \mathbb{R})}$. One crucial point is that for any $a \in [0, 1)$, $P^{\text{gas}}(*)$ contains the element $\sum_{n \in \mathbb{N}} a^n \cdot T^n$.

Proof. Consider the null sequence $(1, T, T^2, ...)$ in $P = \mathbb{Z}[\hat{T}]$, we specify $q = \sqrt[n]{\frac{1}{2}} > a$ for some n and let $b := \frac{a}{q} < 1$. Consider $(1, b, b^2, ...)$ a null sequence in $\mathbb{R}$ and $(1, bT, b^2T^2, ...)$ is a null sequence in $\mathbb{Z}[\hat{T}] \otimes \mathbb{R}$ which maps to P^{gas}. Since Id $-q \cdot$ shift acts as an isomorphism, we know $\sum_{n=0}^{\infty} b^n T^n q^n$ exists and by setup, it is the same as $\sum_{n=0}^{\infty} a^n \cdot T^n$. ■

Now fix $1 < r' < r$, denote by a the ratio $\frac{r'}{r}$ and let $0 < b, c < 1$ be such that $a = bc$. We have

$$\begin{tikzcd}
A_r(T) = P^{\mathrm{gas},\vee} \arrow[rr, "(\cdot a^n)_{n\in\mathbb{N}}"] \arrow[dr, "(\cdot b^n)_{n\in\mathbb{N}}"'] & & A_{r'}(T) = P^{\mathrm{gas},\vee} \\
& P^{\mathrm{gas}} \arrow[ur, "(\cdot c^n)_{n\in\mathbb{N}}"'] &
\end{tikzcd}$$

We write $P^{\mathrm{gas},\vee}$ for $\underline{\mathrm{Hom}}_{\mathbb{R}}(P^{\mathrm{gas}}, \underline{\mathbb{R}}) \cong \underline{C_0(\mathbb{N}, \mathbb{R})}$. Then the map $(\cdot a^n)_{n\in\mathbb{N}}$ is given above as the injective transition map. The map $(\cdot c^n)_{n\in\mathbb{N}} \in \mathrm{Hom}_{\mathbb{R}_{gas}}(P^{\mathrm{gas}}, P^{\mathrm{gas},\vee}) = \mathrm{Hom}_{\mathbb{R}}(P^{\mathrm{gas}} \otimes_{\mathbb{R}^{\mathrm{gas}}} P^{\mathrm{gas}}, \mathbb{R}) = \mathrm{Hom}_{\mathbb{Z}}(P \otimes P, \mathbb{R}) = C_0(\mathbb{N} \times \mathbb{N}, \mathbb{R})$. Thus we can see $(\cdot c^n)_{n\in\mathbb{N}}$ is the particular map take $(n, m) \mapsto \delta_{n,m} c^n$. Finally the map $(\cdot b^n)_{n\in\mathbb{N}} : P^{\mathrm{gas},\vee} \to P^{\mathrm{gas}}$ factors as $P^{\mathrm{gas},\vee} \xrightarrow{\mathrm{Id}\otimes\alpha} P^{\mathrm{gas},\vee} \otimes_{\mathbb{R}^{\mathrm{gas}}} P^{\mathrm{gas}} \otimes_{\mathbb{R}^{\mathrm{gas}}} P^{\mathrm{gas}} \xrightarrow{\mathrm{ev}\otimes\mathrm{Id}} P^{\mathrm{gas}}$ (everything is underived thanks to the internal projectivity of P). The map $\alpha : \mathbb{R} \to P^{\mathrm{gas}} \otimes_{\mathbb{R}^{\mathrm{gas}}} P^{\mathrm{gas}}$ is given by the composition of the diagonal map on P^{gas} with $\mathbb{R} \to P^{\mathrm{gas}} : 1 \mapsto \sum_{n=0}^{\infty} b_n \cdot T^n$. We showed previously the image of 1 indeed lies in $P^{\mathrm{gas}}(*)$.

Theorem 1.30 (Clausen-Scholze)**.** *For $S \in \mathrm{Prof}_{\mathbb{N}}$, $\mathbb{R}[S]^{\mathrm{gas}}$ is quasi-separated. Moreover,*

$$\mathbb{R}[\hat{T}]^{\mathrm{gas}}(*) = \left\{ \sum_{n=0}^{\infty} x_n \cdot T^n \;\middle|\; \begin{array}{c} x_n \text{ have quasi-exponential decay, i.e.} \\ \exists \epsilon > 0, c > 0 \text{ such that } |x_n| \leqslant c \cdot (\frac{1}{2})^{n^\epsilon} \end{array} \right\}.$$

To better understand this theorem, we can look at the gaseous ring structure of $\mathbb{Z}(\!(q)\!)$ (which is motivated by defining the Tate curve). More precisely, consider $R_0 = (R_0^{\triangleright}, \mathrm{Mod}_{R_0}^{\mathrm{gas}})$, where $R_0^{\triangleright}$ is the subring $\mathbb{Z}[\hat{q}][\frac{1}{q}] \subseteq \mathbb{Z}(\!(q)\!)$ and Mod_{R_0} is defined to be q-gaseous $R_0^{\triangleright}$-modules. Then R_0 is a pre-analytic ring but not analytic one. Let us denote the analytic completion of R_0 by $R = (R^{\triangleright}, \mathrm{Mod}_R)$. We have

$$\begin{aligned}
R^{\triangleright} &= \bigcup_{\substack{k>0 \\ n>0}} \prod_{m\geqslant -n} \left(\mathbb{Z} \cap [-(m+n)^k, (m+n)^k]\right) q^m \\
R[S] &= \left\{ \mu = \sum_{m\in\mathbb{Z}} \mu_m q^m \in \mathbb{Z}(\!(q)\!)[S]^{\square} \;\middle|\; \begin{array}{c} \mu_m \in \mathbb{Z}[S] \subseteq \mathbb{Z}[S]^{\square}, \\ |\mu_n| \text{ has at most polynomial growth} \end{array} \right\} \\
&= \bigcup_{\substack{k>0 \\ n>0}} \varprojlim_i \left(\prod_{m\geqslant -n} \mathbb{Z}[S_i]_{\ell^1 \leqslant (m+n)^k} q^m \right)
\end{aligned}$$

Proposition 1.31. *There is an isomorphism $\mathbb{R}^{\mathrm{gas}} \cong R/(1-2q) = (\mathbb{Z}[\hat{q}][\frac{1}{q}], \mathrm{Mod}^{\mathrm{gas}})^{\wedge}/(1-2q)$.*

In particular, we can see $\mathbb{R}[S]^{\mathrm{gas}} = \varinjlim_{k>0,c>0} \varprojlim_i \left\{ \sum_{s\in S_i} x_s[s] \in \mathbb{R}[S_i] \,\middle|\, \sum_{s\in S_i} f_k(x_s) \leqslant c \right\}$, here we take f_k to be any increasing concave function so that for $|x|$ small, $f_k(x) = |\ln(|x|)|^{-k}$. The condition can be translated back to the form in the Theorem 1.30 above.

2 Analytic Stacks

2.1 Heuristic examples

From now on, we 'upgrade' everything into derived/higher categorical setup. Namely we should think of analytic rings as pair $(A^{\triangleright}, \mathrm{D}_{\geqslant 0}(A))$ where $A^{\triangleright}$ is an animated ring and $\mathrm{D}(A)$ is the connective ∞-derived full subcategory of $\mathrm{D}(A^{\triangleright})$ so that the analytic completion functor $\mathrm{D}(A^{\triangleright}) \to \mathrm{D}(A)$ restricts to the connective objects.

Example 2.1. (Solid affine line) Our first motivating example is the affine line in solid geometry. We define $\mathbb{Z}[T]_{\square} = (\mathbb{Z}[T], \mathrm{D}_{\geqslant 0}(\mathbb{Z}[T]_{\square}))$, where the later $\mathrm{D}_{\geqslant 0}(\mathbb{Z}[T]_{\square})$ is the T-solidification:

$$\mathrm{D}_{\geqslant 0}(\mathbb{Z}[T]_{\square}) = \left\{ M \in \mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{D}_{\geqslant 0}(\mathbb{Z}_{\square})) \,\middle|\, R\underline{\mathrm{Hom}}_{\mathbb{Z}}(P, M) \xrightarrow[\cong]{1-T\cdot\mathrm{shift}} R\underline{\mathrm{Hom}}_{\mathbb{Z}}(P, M) \right\}$$

Notice the condition on M is equivalent to

$$\begin{aligned} & R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}(P \otimes \mathbb{Z}[T], M) \xrightarrow[\cong]{1-T\cdot\mathrm{shift}} R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}(P \otimes \mathbb{Z}[T], M) \\ \Leftrightarrow & R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]} \left((P \otimes_{\mathbb{Z}} \mathbb{Z}[T])^{\square} / (1 - T \cdot \mathrm{shift}), M \right) = 0 \end{aligned}$$

We have also $(P \otimes_{\mathbb{Z}} \mathbb{Z}[T])^{\square} = P^{\square} \otimes_{\mathrm{Solid}} \mathbb{Z}[T] = \mathbb{Z}[\![U]\!][T]$ and thus

$$(P \otimes_{\mathbb{Z}} \mathbb{Z}[T])^{\square} / (1 - T \cdot \mathrm{shift}) = (P \otimes_{\mathbb{Z}} \mathbb{Z}[T])^{\square} / (1 - TU) = \mathbb{Z}(\!(T^{-1})\!)$$

The requirement for $M \in \mathrm{D}_{\geqslant 0}(\mathbb{Z}[T]_{\square})$ turns into $R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}(\mathbb{Z}(\!(T^{-1})\!), M) = 0$ and we can think of $\mathrm{D}_{\geqslant 0}(\mathbb{Z}[T]_{\square})$ as localizing $\left(\mathbb{Z}[T], \mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{D}_{\geqslant 0}(\mathbb{Z}_{\square}))\right)$ with respect to $\mathbb{Z}(\!(T^{-1})\!)$. Ideally the 'analytic spectrum' AnSpec which assigns analytic rings to 'affine analytic spaces' should be applied to analytic rings above and gives us:

$$\begin{tikzcd} (\mathbb{A}^{1,\square})^c \arrow[r, hook, "i"] \arrow[d, phantom, "\shortparallel"] & \mathbb{A}^{1,\mathrm{alg}} \arrow[d, phantom, "\shortparallel"] & \mathbb{A}^{1,\square} \arrow[l, hook', "j"'] \arrow[d, phantom, "\shortparallel"] \\ \mathrm{AnSpec}\left((\mathbb{Z}(\!(T^{-1})\!), \mathbb{Z})_{\square}\right) & \mathrm{AnSpec}\left((\mathbb{Z}[T], \mathbb{Z})_{\square}\right) & \mathrm{AnSpec}\left(\mathbb{Z}[T]_{\square}\right) \end{tikzcd}$$

In general for $A^{\triangleright} \to B^{\triangleright}$ and analytic ring $A = (A^{\triangleright}, \mathrm{D}_{\geqslant 0}(A))$, we use $B_{A/}$ (over category) to denote the induced analytic structure for $\left(B^{\triangleright}, A \otimes_{A^{\triangleright}} B^{\triangleright} = \mathrm{Mod}_{B^{\triangleright}}(\mathrm{D}_{\geqslant 0}(A))\right)$. In above diagram $(\mathbb{Z}(\!(T^{-1})\!), \mathbb{Z})_{\square}$ resp. $(\mathbb{Z}[T], \mathbb{Z})_{\square}$ are the induced analytic rings for $\mathbb{Z}(\!(T^{-1})\!)_{\mathbb{Z}_{\square}/}$ resp. $\mathbb{Z}[T]_{\mathbb{Z}_{\square}/}$. We would like to capture morphisms i, j via six functors. We can consider the following diagram:

$$\begin{tikzcd} \mathrm{D}_{\geqslant 0}\left((\mathbb{Z}(\!(T^{-1})\!), \mathbb{Z})_{\square}\right) \arrow[r, "i_*"] & \mathrm{D}_{\geqslant 0}\left((\mathbb{Z}[T], \mathbb{Z})_{\square}\right) \arrow[l, bend right, "i^!"'] \arrow[l, bend left, "i^*"] \arrow[r, "j^*"] & \mathrm{D}_{\geqslant 0}\left(\mathbb{Z}[T]_{\square}\right) \arrow[l, bend right, "j_*"'] \arrow[l, bend left, "j_!"] \end{tikzcd}$$

Here i_* and j_* are taken to be the natural forgetful functor. Let us first take a look on the left side. We take i^* to be $\mathbb{Z}(\!(T^{-1})\!) \otimes_{(\mathbb{Z}[T],\mathbb{Z})} -$, then it is left adjoint of i_* by definition. This adjointness is expected to hold as the notations we take. Likewise, we put $i^! := R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}(\mathbb{Z}(\!(T^{-1})\!), -)$, then $i_* \dashv i^!$ and we put $i_! = i_*$. Now we take a look on the right side of the diagram. The functor j^* is taken to be the T-solidification, it is left adjoint to j_*. The functor $j_!$ is more mysterious, we postpone it to later discussion.

Lemma 2.2. *The composition $j^* \circ i_* = j^* \circ i_! = 0$.*

Proof. For any $M \in \mathrm{D}\left((\mathbb{Z}(\!(T^{-1})\!), \mathbb{Z})_\square\right) = \mathrm{Mod}_{\mathbb{Z}(\!(T^{-1})\!)}(\mathrm{D}(\mathbb{Z}_\square))$, we want to show $j^* i_!(M) = j^*(M) = 0 \Leftrightarrow \forall N \in D(\mathbb{Z}[T]_\square), R\,\mathrm{Hom}_{\mathbb{Z}[T]_\square}(j^* M, N) = R\,\mathrm{Hom}_{(\mathbb{Z}[T],\mathbb{Z})_\square}(M, N) = 0$. For some nonzero $N \in D(\mathbb{Z}[T]_\square)$, we can consider the orthogonal category of $\mathcal{C} := \mathrm{Mod}_{\mathbb{Z}[T]}(\mathrm{D}(\mathbb{Z}_\square))$ to be $\mathcal{C}^{\perp N} := \{M \in \mathcal{C} \mid R\underline{\mathrm{Hom}}(M, N) = 0\}$.
We have: (1) $\mathbb{Z}(\!(T^{-1})\!) \in \mathcal{C}^{\perp N}$; (2) $\forall M' \in \mathcal{C}$, $\mathbb{Z}(\!(T^{-1})\!) \otimes M' \in \mathcal{C}^{\perp N}$; (3) $\mathcal{C}^{\perp N}$ is closed under colimit.
In fact, (1) follows from the definition of $\mathrm{D}(\mathbb{Z}[T]_\square)$ and (2) follows from (1) by adjointness $\overset{\mathbb{L}}{\otimes} \dashv R\underline{\mathrm{Hom}}$. For $(M_i) \in \mathcal{C}^{\perp N}$, we notice that $R\underline{\mathrm{Hom}}(\varinjlim_i M_i, N) = \varprojlim_i R\underline{\mathrm{Hom}}(M_i, N) = 0$. With (1) to (3) above, one can deduce that $M \in \mathrm{D}\left((\mathbb{Z}(\!(T^{-1})\!), \mathbb{Z})_\square\right) \subseteq \mathcal{C}^{\perp N}$. ■

By definition, we have $j^* M \in \mathcal{C}^{\perp \mathbb{Z}(\!(T^{-1})\!)}, \forall M \in \mathcal{C}$. Let $I = (\mathbb{Z}(\!(T^{-1})\!)/\mathbb{Z}[T])[-1] = \mathrm{fib}(\mathbb{Z}[T] \to \mathbb{Z}(\!(T^{-1})\!))$, we claim that j^* can be explicitly computed by $R\underline{\mathrm{Hom}}(I, -)$. Apply $R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}(-, M)$ and j^* to I we get:

$$j^* R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}(\mathbb{Z}(\!(T^{-1})\!), M) \to j^* M \to j^* R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}(I, M) \xrightarrow{+1}$$

By previous Lemma we get $j^* R\underline{\mathrm{Hom}}_{\mathbb{Z}[T]}(\mathbb{Z}(\!(T^{-1})\!), M) = 0$, thus our goal is to show

$$R\underline{\mathrm{Hom}}\left(\mathbb{Z}(\!(T^{-1})\!), R\underline{\mathrm{Hom}}(I, M)\right) = 0 \Leftrightarrow R\underline{\mathrm{Hom}}(\mathbb{Z}(\!(T^{-1})\!) \overset{\mathbb{L}}{\otimes} I, M) = 0$$

It is enough to show $\mathbb{Z}(\!(T^{-1})\!) \overset{\mathbb{L}}{\otimes} I = 0 \Leftrightarrow \mathbb{Z}(\!(T^{-1})\!) \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}[T]} \mathbb{Z}(\!(T^{-1})\!) \cong \mathbb{Z}(\!(T^{-1})\!)$. We have

$$\mathbb{Z}(\!(T^{-1})\!) \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}[T]} \mathbb{Z}(\!(T^{-1})\!) = \left(\mathbb{Z}(\!(T^{-1})\!) \overset{\mathbb{L}}{\otimes}_{\mathbb{Z}_\square} \mathbb{Z}(\!(T^{-1})\!)\right) \Big/^{\mathbb{L}} (T \otimes 1 - 1 \otimes T) \cong \mathbb{Z}(\!(T_1^{-1}, T_2^{-1})\!)/(T_1 - T_2) \cong \mathbb{Z}(\!(T^{-1})\!) \quad ■$$

We can see the idempotency of $\mathbb{Z}(\!(T^{-1})\!)$ matters. Use the notation above $I := (\mathbb{Z}(\!(T^{-1})\!)/\mathbb{Z}[T])[-1]$, we complete the above diagram by setting $j_! := I \overset{\mathbb{L}}{\otimes} -$, then $j_! \dashv j^*$. From point of view of adjoint functors, we can regard j as an open immersion while i as a closed embedding.

The above construction can be generalized. Let $\mathcal{C}$ be a unital symmetric monoidal category with internal Hom. Let $A \in \mathrm{CAlg}(\mathcal{C})$ be a commutative ring object in $\mathcal{C}$. We first require A to be idempotent

$$\begin{tikzcd} \mathrm{Mod}_A(\mathcal{C}) \arrow[r, "i_*"] & \mathcal{C} \arrow[l, bend right, "i^*"'] \arrow[l, bend left, "i^!"] \arrow[r, "j^*"] & \mathcal{C}^{\perp A} \arrow[l, bend right, "j_*"'] \arrow[l, bend left, "j_!"] \end{tikzcd} \tag{2.1.1}$$

Similar to the above, in the above diagram, we define $\mathcal{C}^{\perp A} := \{M \in \mathcal{C} \,|\, \underline{\mathrm{Hom}}(A, M) = 0\}$. Also we put $i_* = i_!$ and j^* be the forgetful functors, $i^* = A \otimes -$, $i^! = \underline{\mathrm{Hom}}(A, -)$ and $j^* = j^! := \underline{\mathrm{Hom}}(I, -)$, where $I := \mathrm{fib}(\mathbf{1}_{\mathcal{C}} \to A)$.

Lemma 2.3. *We have $j^*\underline{\mathrm{Hom}}(A, M) = 0, \forall M \in \mathcal{C}$ and j^*M is indeed in $\mathcal{C}^{\perp A}$*

Proof. By construction, $j^*\underline{\mathrm{Hom}}(A, M) = 0 \Leftrightarrow \underline{\mathrm{Hom}}(A, M) = \underline{\mathrm{Hom}}(A \otimes A, M)$, which is obvious since A is taken to be idempotent. Since $\mathcal{C}$ is symmetric, we also have $\underline{\mathrm{Hom}}(A, \underline{\mathrm{Hom}}(I, M)) = \underline{\mathrm{Hom}}(I, \underline{\mathrm{Hom}}(A, M)) = 0$. ■

In this case, we can put $j_! := I \otimes -$ and we have the adjunctions $i^! \dashv i_! \dashv i^*$ and $j_! \dashv j^! \dashv j_*$.
For general $A \in \mathrm{CAlg}(\mathcal{C})$, j^* is not defined as well as $j_!$. At least we want $j^*(M)$ still lands in $\mathcal{C}^{\perp A}$ and j^* is a left adjoint of the inclusion functor j_*.

Proposition 2.4. *We consider the following cases either (i) $\underline{\mathrm{Hom}}(I^{\otimes n}, M)$ stabilizes for $n >> 0$ and for all M. or (ii) $\underline{\mathrm{Hom}}(A, -)$ commutes with sequential limits.*
Then $j^*(M) := \varinjlim_n \underline{\mathrm{Hom}}(I^{\otimes n}, M)$ lands in $\mathcal{C}^{\perp A}$ and j^* is left adjoint to j_*.

Proof. In both of the above two situations, we have

$$\underline{\mathrm{Hom}}(A, \varinjlim_n \underline{\mathrm{Hom}}(I^{\otimes n}, M)) \cong \varinjlim_n \underline{\mathrm{Hom}}(A, \underline{\mathrm{Hom}}(I^{\otimes n}, M))$$

Also we have $\varinjlim_n \underline{\mathrm{Hom}}(I, \underline{\mathrm{Hom}}(I^{\otimes n}, M)) = \varinjlim_n \underline{\mathrm{Hom}}(I^{\otimes n+1}, M) = \varinjlim_n \underline{\mathrm{Hom}}(I^{\otimes n}, M)$. Since

$$\underline{\mathrm{Hom}}(A, \underline{\mathrm{Hom}}(I^{\otimes n}, M)) \to \underline{\mathrm{Hom}}(I^{\otimes n}, M) \to \underline{\mathrm{Hom}}(I^{\otimes n+1}, M)$$

Take the colimit we find $\varinjlim_n \underline{\mathrm{Hom}}(A, \underline{\mathrm{Hom}}(I^{\otimes n}, M)) = 0$, thus $\varinjlim_n \underline{\mathrm{Hom}}(I^{\otimes n}, M) \in \mathcal{C}^{\perp A}$. ■

Example 2.5. (i) Take $\mathcal{C} = \mathrm{D}_{\geqslant 0}((\mathbb{Z}[T], \mathbb{Z})_{\square})$ and $A = \mathbb{Z}(\!(T^{-1})\!)$, we can recover the previous example.
(ii) Take $\mathcal{C} = \mathrm{D}_{\geqslant 0}(\mathbb{Z})$ and $A = P/(1-q)$, where we write $P = \mathbb{Z}[\hat{q}]$, then $\mathcal{C}^{\perp A} = \mathrm{D}_{\geqslant 0}(\mathrm{Solid})$.

Example 2.6 (Adic unit disc)**.** We consider the Tate algebra $\mathbb{Q}_p\langle T\rangle$ and its subring of integral elements $\mathbb{Z}_p\langle T\rangle$.

Lemma 2.7. *The T-solidification of $\mathbb{Z}[T] \otimes_{\mathbb{Z}_\square} \mathbb{Z}_{p,\square}$ is isomorphic to $\mathbb{Z}_p\langle T\rangle$, where $\mathbb{Z}_{p,\square}$ is defined in 1.24 (iii).*

Proof. We view $\mathbb{Z}[T] \otimes_{\mathbb{Z}_\square} \mathbb{Z}_{p,\square} = \mathrm{Mod}_{\mathbb{Z}[T]}(\mathbb{Z}_{p,\square}) \subseteq \mathrm{Mod}_{\mathbb{Z}[T]}(\mathbb{Z}_\square)$. Thus the T-solidification, use the previous notation, is applying the functor j^*. We claim that there is a $\mathbb{Z}(\!(T^{-1})\!)$-module structure on $\mathbb{Z}_p\langle T\rangle/\mathbb{Z}_p[T]$. In fact the action of $\mathbb{Z}(\!(T^{-1})\!)$ on $\mathbb{Z}_p\langle T\rangle/\mathbb{Z}_p[T]$ is given by $(\sum a_n T^{-n}) \cdot (\sum b_m T^m) := \sum_{m-n\geqslant 0} b_m a_n T^{m-n}$. One can check that this action is well-defined. Hence we observe that $j^*(\mathbb{Z}_p\langle T\rangle/\mathbb{Z}_p[T]) = 0$ and that $j^*\mathbb{Z}_p[T] \xrightarrow{\cong} j^*\mathbb{Z}_p\langle T\rangle \cong \mathbb{Z}_p\langle T\rangle$, the last equality is because $\mathbb{Z}_p\langle T\rangle$ is already T-solid. ■

We can regard the adic unit disc $\mathbb{D}$ as Analytic Spectrum (specified later) of the $(\mathbb{Q}_p\langle T\rangle, \mathbb{Z}_p\langle T\rangle)_\square$, which is also, by above Lemma 2.7, $\left(\mathbb{Q}_p\langle T\rangle, \mathbb{Z}_p[T]\right)_\square$. One can think of the underlying topological space (almost) as the adic Spectrum of Huber pair $(R, R^+) = (\mathbb{Q}_p\langle T\rangle, \mathbb{Z}_p\langle T\rangle)$.
Likewise, one can also formulate the disc of smaller radius $\mathbb{D}(\frac{1}{p^n})$ that associated with the analytic ring

$$\mathrm{D}\left(\left(\mathbb{Q}_p\left\langle\frac{T}{p^n}\right\rangle, \mathbb{Z}_p\left\langle\frac{T}{p^n}\right\rangle\right)_\square\right)$$

Then $\mathbb{D}(\frac{1}{p^n})$ has an inclusion to $\mathbb{D}$, whose completion in $\mathbb{D}$ can be described as $\mathrm{D}\left(\left(A, \mathbb{Z}_p\langle T\rangle\right)_\square\right)$, where A is defined to be

$$A = \mathbb{Z}_p(\!((\frac{T}{p^n})^{-1})\!)^{T-\square}[\frac{1}{p}] \in \mathrm{D}\left(\mathbb{Q}_p\langle T\rangle, \mathbb{Z}_p[T]\right)_\square$$

Another interesting case we can consider is the dagger (overconvergent) space $\{0\}^\dagger$, which denotes

$$\{0\}^\dagger := \mathrm{AnSpec}\left(\varinjlim_n \mathbb{Q}_p\left\langle \frac{T}{p^n}\right\rangle, \mathbb{Z}_p\langle T\rangle\right)_\square$$

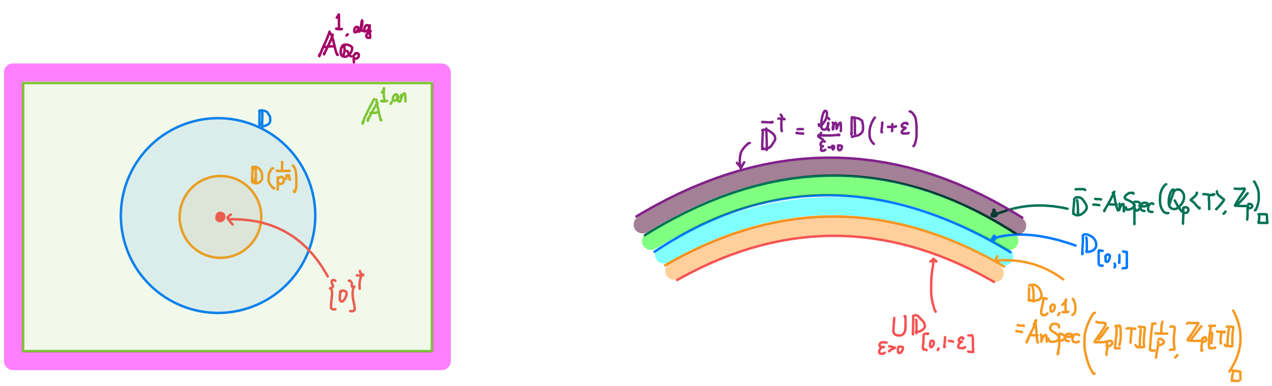
We can view $\{0\}^\dagger$ as closed subspace in $\mathbb{D}$ due to the following Lemma:

Lemma 2.8. *$\varinjlim_n \mathbb{Q}_p\left\langle \frac{T}{p^n}\right\rangle$ is an idempotent algebra.*

Proof. It is enough to investigate for $\mathbb{Z}_p\left\langle \frac{T}{p^n}\right\rangle$. Over $\mathbb{Z}_p\langle T\rangle$, we have:

$$\mathbb{Z}_p\left\langle \frac{T}{p^n}\right\rangle \otimes_{\mathbb{Z}_p\langle T\rangle} \mathbb{Z}_p\left\langle \frac{T}{p^n}\right\rangle \cong \left(\mathbb{Z}_p\left\langle \frac{T_1}{p^n}, \frac{T_2}{p^n}\right\rangle/(T_1-T_2)\right)^{T-\square} \quad \text{Let } U := \frac{T_1-T_2}{p^n}$$
$$\cong \left(\mathbb{Z}_p\left\langle \frac{T}{p^n}, U\right\rangle/p^nU\right)^{T-\square} \cong \left(\mathbb{Z}_p\left\langle \frac{T}{p^n}\right\rangle[U]/p^nU\right)^{T-\square} \cong \mathbb{Z}_p\left\langle \frac{T}{p^n}\right\rangle[U]/p^nU$$

Taking $[\frac{1}{p}]$ and the colimit we get what we want. ∎



(a) Inclusions inside affine line

(b) Inclusions among disks

Figure 1

The above Figure 1a and 1b give us a rough illustration how the inclusions are like for analytic spaces in dimension one. We can describe the spaces appear in the above pictures by following sequence:

$$\{0\}^\dagger \subseteq \mathbb{D}_{[0,1^-]} = \bigcup_{\epsilon>0} \mathbb{D}_{[0,1-\epsilon]} \subseteq \mathbb{D}_{[0,1)} \subseteq \mathbb{D} = \mathbb{D}_{[0,1]} \subseteq \bar{\mathbb{D}} \subseteq \bar{\mathbb{D}}^\dagger = \varprojlim_{\epsilon\to 0} \mathbb{D}(1+\epsilon) \subseteq \mathbb{A}^{1,\mathrm{an}}_{\mathbb{Q}_p} \subseteq \mathbb{A}^{1,\mathrm{alg}}_{\mathbb{Q}_p} \subseteq \mathbb{P}^{1,\mathrm{alg}}$$

Here $\mathbb{D}_{[0,1)}$ is can be regard as the analytic spectrum $\mathrm{AnSpec}\left(\mathbb{Z}_p[\![T]\!][\frac{1}{p}], \mathbb{Z}_p[\![T]\!]\right)_\square$ and we can notice it is

NOT a Huber pair. As is depicted in Figure 1b, $\mathbb{D}_{[0,1)} \subseteq \mathbb{D}$, topologically the points missing are rank 2 points $\left\{x_1^-\right\} \subseteq \operatorname{Spa}(\mathbb{Q}_p\langle T\rangle, \mathbb{Z}_p\langle T\rangle)$. Likewise, we have $\mathbb{D} \subseteq \bar{\mathbb{D}}$, the latter is defined as $\operatorname{AnSpec}(\mathbb{Q}_p\langle T\rangle, \mathbb{Z}_p)_\square$ and topologically the points added into $\mathbb{D}$ are rank 2 points $\left\{x_1^+\right\}$. We can regard $\bar{\mathbb{D}}$ as a compactified unit disk. And $\mathbb{D}_{[0,1^-]}$, the union of smaller disks, could not be expressed as a single affinoid.

Example 2.9 (Zariski localization)**.** Let R be (animated) ring and $f \in R$, we can think of the diagram 2.1.1 as (classical) localizing. Put $A = R[\frac{1}{f}]$, then we have $\operatorname{Mod}_A(\mathrm{D}(R)) = \mathrm{D}(R[\frac{1}{f}])$ and $\mathrm{D}(R)^{\perp A}$ can be written as

$$\mathrm{D}(R)^{\perp A} = \left\{ M \in \mathrm{D}(R) \,\middle|\, R\underline{\operatorname{Hom}}(R[\frac{1}{f}], M) = 0 \right\} \Leftrightarrow M \cong \varprojlim_n \left(M \big/^{\mathbb{L}} f^n \right)$$

namely $M \in \mathrm{D}(R)^{\perp A}$ if it is f-adically complete. Similar to the previous construction, we let $j^* : \mathrm{D}(R) \to \mathrm{D}(R)^{\perp A}$ be the functor $R\underline{\operatorname{Hom}}(R[\frac{1}{f}]/^{\mathbb{L}}R, -)[1]$. Then $j^*(M) = \varprojlim_n \left(M \big/^{\mathbb{L}} f^n \right)$ is the f-adic completion functor. Put $j_! = R[\frac{1}{f}]/^{\mathbb{L}}R \otimes -[1]$, it is left adjoint to j^*. So we get the following diagram:

$$\begin{tikzcd}
\mathrm{D}(R[\frac{1}{f}]) \arrow[r, "i_*"] & \mathrm{D}(R) \arrow[l, bend right, "i^*"'] \arrow[l, bend left, "i^!"] \arrow[r, "j^*"] & \mathrm{D}(\varprojlim R/^{\mathbb{L}} f^n) \arrow[l, bend right, "j_*"'] \arrow[l, bend left, "j_!"]
\end{tikzcd}$$

For the Zariski topology of ring spectrum, we have $\operatorname{Spec}(R[\frac{1}{f}]) \xhookrightarrow{\text{open}} \operatorname{Spec} R$ which is open. Under the philosophy of locales, i_* should be regarded as something like "closed map".
If we take $A := R/^{\mathbb{L}}f$, then $\mathrm{D}(R)^{\perp A} = \left\{ M \in \mathrm{D}(R) \,\middle|\, R\underline{\operatorname{Hom}}(R/^{\mathbb{L}}f, M) = 0 \right\} = \mathrm{D}(R[\frac{1}{f}])$. And the functor $j^* : \mathrm{D}(R) \to \mathrm{D}(R)^{\perp A}$ is given by $j^*M = \varinjlim_{\times f} M = M[\frac{1}{f}]$.

Example 2.10 (Gaseous case)**.** We now explain more explicitly the gaseous structure of $R_0^{\triangleright} = \mathbb{Z}[\hat{q}][q^{-1}]$ that appears in Proposition 1.31. The computation reduces to the case for $R_0^{\triangleright} = \mathbb{Z}[\hat{q}]$, we have

Theorem 2.11. *Let $S = \varprojlim_i S_i \in \operatorname{Prof}_{\mathbb{N}}$, then*

$$\mathbb{Z}[\hat{q}]^{\mathrm{gas}}[S] := \left(\mathbb{Z}[\hat{q}][S]\right)^{\mathrm{gas}} \cong \varinjlim_{n,k} \varprojlim_i \left(\prod_{m \geqslant 0} \mathbb{Z}[S_i]_{\ell^1 \leqslant (m+n)^k} q^m \right)$$

In particular, if we take $S = *$, then $\mathbb{Z}[\hat{q}]^{\mathrm{gas}} = \varinjlim_{n,k} \prod_m \mathbb{Z}_{\mathrm{abs} \leqslant (m+n)^k} q^m = \left\{ \sum_m a_m \cdot q^m \,|\, |a_m| \text{ has polynomial growth} \right\}$, as we have seen. Let $A := \mathbb{Z}[\hat{q}] \otimes P/(1-q\text{-shift})$ and $I = \operatorname{fib}(\mathbb{Z}[\hat{q}] \to A) = \operatorname{cofib}\left(\mathbb{Z}[\hat{q}] \oplus (\mathbb{Z}[\hat{q}] \otimes P) \to \mathbb{Z}[\hat{q}] \otimes P\right)[-1]$. We consider the localization of R_0 with respect to A, let $P = \mathbb{Z}[\hat{y}]$ so that $\cdot$ shift $= \cdot y^{-1}$ and $M := \mathbb{Z}[\hat{q}][S]$, then we have:

$$R\underline{\operatorname{Hom}}(P, \mathbb{Z}[S]) = \varinjlim_n \varprojlim_i \left(\mathbb{Z}[S_i]_{\ell^1 \leqslant n}[\hat{y}] \right);$$

$$R\underline{\operatorname{Hom}}(P, \mathbb{Z}[\hat{q}][S]) = \varinjlim_n \varprojlim_{m,i} ((\mathbb{Z}[q]/q^m)[S_i])_{\ell^1 \leqslant n} [\hat{y}].$$

And thus

$$
\begin{aligned}
R\underline{\mathrm{Hom}}_{\mathbb{Z}[\hat{q}]}(I,M) &= \mathrm{fib}\left(R\underline{\mathrm{Hom}}(P,M) \xrightarrow{1-qy^{-1}} R\underline{\mathrm{Hom}}(P,M)\oplus M\right)[1] \\
&= \mathrm{fib}\left(\varinjlim_n \varprojlim_{m,i} ((\mathbb{Z}[q]/q^m)[S_i])_{\ell^1\leqslant n}[\hat{y}] \xrightarrow{\begin{pmatrix} 1-q\cdot y^{-1} \\ y=0 \end{pmatrix}} \begin{matrix} \varinjlim_n \varprojlim_{m,i} ((\mathbb{Z}[q]/q^m)[S_i])_{\ell^1\leqslant n}[\hat{y}] \\ \oplus \\ \varinjlim_n \varprojlim_{m,i} ((\mathbb{Z}[q]/q^m)[S_i])_{\ell^1\leqslant n}[\hat{y}] \end{matrix} \right)[1] \\
&= \varinjlim_n \varprojlim_{m,i} ((\mathbb{Z}[q]/q^m)[S_i])_{\ell^1\leqslant n}[y] \Big/^{\mathbb{L}} (q-y)
\end{aligned}
$$

In general we can calculate $R\underline{\mathrm{Hom}}\left(I^{\otimes N}, \mathbb{Z}[\hat{q}][S]\right)$, which is $\varinjlim_n \varprojlim_{m,i} ((\mathbb{Z}[q]/q^m)[S_i])_{\ell^1\leqslant n}[y_1,\ldots,y_N]\Big/^{\mathbb{L}}(q-y_1,\ldots,q-y_N)$. Moreover, it admits an embedding into $\mathbb{Z}^{\square}[S][[q]] = \varprojlim_{m,i} \mathbb{Z}[S_i][q]\Big/q^m$ by sending $y_i \mapsto q$. In fact, we can see for each piece $\varprojlim_m (\mathbb{Z}[q]/q^m)_{\ell^1\leqslant n}[y_1,\ldots,y_N]$, the coefficient of q^m with respect to the embedding $R\underline{\mathrm{Hom}}\left(I^{\otimes N}, \mathbb{Z}[\hat{q}]\right) \hookrightarrow \mathbb{Z}^{\square}[[q]]$ is given by linear combination of $a_{\underline{m}} q^{m_0} y_1^{m_1}\ldots y_N^{m_N} \mapsto a_{\underline{m}} q^{m_0+m_1+\ldots+m_N} = a_{\underline{m}} q^m$, where $\underline{m} = (m_0,\ldots,m_N)$ is a partition of m, $a_{\underline{m}}$ has norm $\leqslant n$ and the number of choices is $\binom{m+N-1}{N}$.
Unlike the previous examples in which the fiber of localizing I is always idempotent. Here in gaseous case, I is not idempotent, however the situation of Proposition 2.4 still holds.

2.2 Six functor formalism

We want to realize analytic stacks as sheaves of ∞-categories over analytic rings, with Grothendieck topology. Moreover, the association $X \mapsto \mathrm{D}(X)$ should come up with six functor formalism. Namely each $\mathrm{D}(X)$ should be equipped with symmetric monoidal structure with internal Hom and for $f : X \to Y$, we want adjoint pairs

$$f^* \dashv f_*; \quad f_! \dashv f^!; \quad \text{and } \otimes \dashv \underline{\mathrm{Hom}}$$

Also we require the proper base change: for commutative diagram

$$
\begin{tikzcd}
X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\
Y' \arrow[r, "g"'] & Y
\end{tikzcd}
$$

We need $g^* f_! \cong f'_! g'^*$. However the point is a priori there is no natural map between those two. In classical algebraic geometry, we have the pull-back functor f^* and its right adjoint f_*. When the morphism is proper then $f_!$ can be set to be f_*. Back to the above commutative diagram, we have a natrual map $\mathrm{Id} \to g'_* g'^*$, thus $f_* \to f_* g'_* g'^* = g_* f'_* g'^*$ and we get a natural map $g^* f_* \to f'_* g'^*$. In this case we can also pin down $f^!$ as the right adjoint of $f_! = f_*$. Similarly if f is an open immersion, we can set $f^! = f_*$ and $f_!$ appears as the left adjoint of $f^!$. Moreover, if $f : X \to Y$ can be decomposed into $X \overset{j}{\hookrightarrow} \bar{X} \xrightarrow{p} Y$ where j is an open immersion

and p is proper, we can define $f_! = p_! \circ j_!$. Namely in that situation, !-maps can be defined, we can define $E = \{\text{!-able maps}\} := \{p \circ j \mid p : \text{proper } j : \text{open}\}$.
Now we back to the case of analytic rings. Let $A = (A^{\triangleright}, \mathrm{D}(A)), B = (B^{\triangleright}, \mathrm{D}(B))$ be two analytic rings with $f : A \to B$ morphism of analytic rings, i.e. it is associated with $f^{\triangleright} : A^{\triangleright} \to B^{\triangleright}$ morphism of condensed animated rings and $f_*^{\triangleright}$ the forgetful functor$\mathrm{D}(B^{\triangleright}) \to \mathrm{D}(A^{\triangleright})$ that restricts to $\mathrm{D}(B) \to \mathrm{D}(A)$. In fact, if $B[S]$ (generator of $\mathrm{D}_{\geqslant 0}(B)$) is also A-complete for all $S \in \mathrm{Prof}_{\mathbb{N}}$, then f is a morphism of analytic rings.
Let f^* be the left adjoint of f_*, for any $M \in \mathrm{D}(A)$, f^*M is the B-completion of $(f^{\triangleright})^*M$.

Definition 2.12. We say f is **proper** if f_* commutes with colimits and satisfies projection formula:

$$M \otimes_A f_* N \xrightarrow{\cong} f_*(f^*M \otimes_B N) \quad \text{(this natural map indeed exists)}$$

Lemma 2.13. *The morphism f of analytic rings is proper if and only if B has an induced analytic structure, i.e. $B = \left(B^{\triangleright}, \mathrm{D}_{\geqslant 0}(B) = \mathrm{Mod}_{B^{\triangleright}}(\mathrm{D}_{\geqslant 0}(A))\right) = A \otimes_{A^{\triangleright}} B^{\triangleright}$.*

Proof. If $B \cong A \otimes_{A^{\triangleright}} B^{\triangleright}$, then it is easy to see the projection formula holds. On the other hand, we can write f^*M as $M \otimes_A B$, which just means the completion of $M \otimes_{A^{\triangleright}} B^{\triangleright}$, so the projection formula says $(M \otimes_A B) \otimes_B N \cong M \otimes_A (N\mid_A)$ in $\mathrm{D}(A)$. Take $N = B^{\triangleright}$, we find $M \otimes_A B = M \otimes_A B^{\triangleright}$ for all $M \in \mathrm{D}(A)$, which also concludes. ■

Definition 2.14. We say f is an **open immersion** if f^* has a fully-faithful left adjoint $f_!$ that satisfies the projection formula.

Lemma 2.15. *If f is an open immersion, then we have $f^* f_* \cong \mathrm{Id} \cong f^* f_!$*

Lemma 2.16. *Fix A an analytic ring, we have bijection between two classes:*

$$\{\textit{open immersion } f : \mathrm{AnSpec}\, B \to \mathrm{AnSpec}\, A\} \cong \left\{ \begin{array}{l} \textit{compact idempotent algebra } C \in \mathrm{D}(A), \\ \textit{such that } R\underline{\mathrm{Hom}}(C/^{\mathbb{L}}A, -)[1] \textit{ preserves } \mathrm{D}_{\geqslant 0}(A) \end{array} \right\}$$

The above correspondence is given by $f \mapsto \mathrm{cofib}(f_! \mathbf{1}_{\mathrm{D}(B)} \to \mathbf{1}_{\mathrm{D}(A)}) =: C_f$.

Proof. We first notice that for f open immersion, C_f in the Lemma is well-defined and $f_!$ satisfies the projection formula by definition. Other properties are OK, we want to see if $C = C_f$ is idempotent, i.e. $C \otimes_A C \cong C$. This is equivalent to $f_! \mathbf{1}_{\mathrm{D}(B)} \otimes_A C \cong 0$. By projection formula, we know it is equivalent to $f_!(\mathbf{1}_{\mathrm{D}(B)} \otimes f^*C) = 0$, thus to $f^*C = 0$, which is true since $f^*C = \mathrm{cofib}(f^* f_! \mathbf{1}_{\mathrm{D}(B)} \to \mathbf{1}_{\mathrm{D}(A)})$ and we can apply Lemma 2.16 above.
Conversely, given C in the right class of the Lemma. We can first associate it with

$$B^{\mathrm{pre}} := \left(A^{\triangleright}, \mathrm{D}_{\geqslant 0}(B) := \left\{ M \in \mathrm{D}_{\geqslant 0}(A) \mid R\underline{\mathrm{Hom}}(C, M) \cong 0 \right\}\right)$$

Notice that $R\underline{\mathrm{Hom}}(C, -)$ commutes with limits (right adjointness) and colimits (since C is compact). We can construct from B^{pre} an analytic ring structure to be $B = \left((A^{\triangleright})^{\wedge}_{B}, \mathrm{D}_{\geqslant 0}(B)\right)$, where the B-completion $(A^{\triangleright})^{\wedge}_{B}$ is nothing but $R\underline{\mathrm{Hom}}(C/^{\mathbb{L}}\mathbf{1}_A, A^{\triangleright})$, by our assumption $(A^{\triangleright})^{\wedge}_{B} \in \mathrm{D}_{\geqslant 0}(A)$ is still connective. ■

Remark 2.17. In the above Lemma, the compactness of C turns out to be important. For example if we take $A = (R, \mathrm{Mod}_R(\mathrm{Solid}))$ and $C = R[\frac{1}{f}]$ the localizing idempotent algebra, which is NOT compact. Then the construction of Lemma gives us a morphism $\varinjlim_n \operatorname{AnSpec} R \Big/^{\mathbb{L}} f^n \longrightarrow \operatorname{AnSpec} C$ (see also Example 2.9). This morphism is open, as we have justified, but it is NOT affine, thus does not fit into the left class in the Lemma.

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